

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CIVIL/EC/ME) Semester – I Examination, 2010
EE - 101 Fundamentals of Electrical & Electronics Engineering

Date: 09.12.2010, Thursday Time: 10:30a.m to 1:30p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary.

SECTION I

- Q-1 A. What is coefficient of coupling? Derive an expression for the same. 5
 B. A series magnetic circuit has an iron path of length 50 cm and an air gap of length 1 mm. The cross-sectional area of the iron is 6.66 cm^2 and the exciting coil has 400 turns. Determine the current required to produce a flux of 0.9 mWb in the air gap. 4

The following points are taken from the magnetisation curve for the iron :

Flux density (T)	1.2	1.35	1.45	1.55
Magnetizing force (AT/m)	500	1000	2000	4000

- C. State and explain Faraday's laws of electromagnetic induction. 3
 Q-2 A. Define average value and obtain expression for a half-wave and full-wave rectified sinusoidal voltage wave. 6
 B. A sinusoidal voltage has a value of 100 volts at 2.5 ms and it takes time of 20 ms to complete one cycle. Find the maximum value and time to reach it for the first time after zero. $e = E_m \sin \omega t \Rightarrow 100 = E_m \sin \omega t = 141.4 \text{ V} = E_m$ $t = 5 \text{ ms}$. 3
 C. Derive the relationship between line voltage and phase voltage, line current and phase current in case of 3 phase star connected system. 3

OR

- Q-2 A. Define RMS value and obtain an expression for a sinusoidal voltage wave. 6
 B. Four alternating currents given by $i_1 = 100 \sin 500t$ $i_3 = -50 \cos 500t$ $i_2 = 200 \sin(500t + \pi/3)$ $i_4 = 150 \sin(500t - \pi/4)$ 3
 are fed into a common conductor. Determine the equation and frequency of the resultant current. $I_x = 306 \text{ A}$ $\phi = 3.2^\circ$ $I_y = 17.2 \text{ A}$ $I_z = 306.48 \text{ A}$ $\tan \phi = 7/1 = 0.056$ $f = 79.58 \text{ Hz}$

- C. Define the following terms: 3
 (i) frequency (ii) form factor (iii) peak factor (iv) time period
 (v) amplitude (vi) instantaneous value
 Q-3 A. State and explain Coulomb's laws of electrostatics. 3
 B. Derive an expression of voltage and current during charging of capacitor. Sketch the variation of voltage and current against time. 5
 C. Calculate the force on a charge of $+10^{-2} \text{ C}$ placed at a point (0, 0) due to point charges Q_1 and Q_2 . The charge $Q_1 = +10^{-6} \text{ C}$ is at (-2, 2) meters and $Q_2 = -2 \times 10^{-6} \text{ C}$ is at (2, 2) meters. 3

$F_1 = 11.25 \text{ N}$ $d = 2\sqrt{2} \text{ m}$
 $F_2 = 22.5 \text{ N}$
 $F_R = 25.156 \text{ N}$

[P.T.O.]

OR

- Q-3 A. Define temperature coefficient of resistance. Derive the following expression incase of good conducting material. Also state its significance. 5

$$\alpha_{t_2} = \frac{1}{\left(\frac{1}{\alpha_{t_1}}\right) + (t_2 - t_1)}$$

- B. A 10 μ F capacitor in series with 1 M Ω resistor is connected across a 100 V d.c. supply. Determine:- 6

- (i) time constant of the circuit 10 sec.
- (ii) the initial value of charging current $V/R = 0.1 \text{ mA}$
- (iii) the initial rate of rise of voltage across the capacitor $V/RC = 10 \text{ V/sec.}$
- (iv) Voltage across the capacitor after a time equal to time constant. $0.632 \times 100 = 63.2 \text{ V}$
- (v) the circuit current at this time $0.368 \times I_m = 0.0368 \text{ mA}$
- (vi) the voltage across the capacitor, 3 seconds after closing the switch 25.92 V
- (vii) the time taken for the capacitor voltage to reach 60 V. 9.16 sec.

SECTION - II

- Q-4 A. Write in brief for following terms: 6

- a. Breakdown voltage
- b. Knee voltage
- c. Forward current
- d. Leakage current
- e. Peak Inverse Voltage
- f. Ripple factor

- B. Explain the operation of biased clipper. 3

- C. A crystal diode having internal resistance $r_f = 10 \Omega$ is used for half-wave rectification. If the applied voltage $v = 25 \sin \omega t$ and load resistance $R_L = 400 \Omega$ find: 3

- a. I_m, I_{dc}, I_{rms} 61 mA, 19.5 mA, 30.5 mA
- b. d.c. output voltage 7.76 V
- c. efficiency of rectification 39.5 %

- Q-5 A. Explain zener diode working as voltage stabilizer. 7

- B. Explain following Logic Gates in detail (i) EX-NOR gate, (ii) AND gate. 2

- C. Give the advantages of digital over analog system 2

OR

- Q-5 A. (I) Explain any three terms 6
(a) duty cycle, (b) fall time, (c) periodic signal, (4) positive logic system

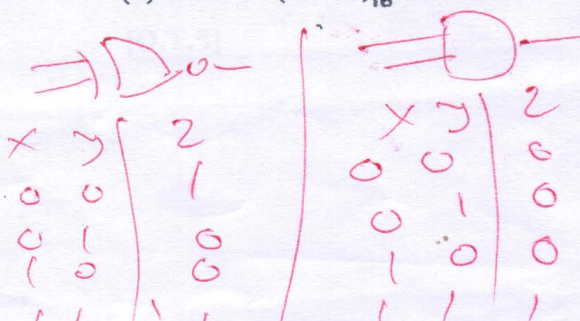
- (II) (a) Perform $(11110)_2 - (10010)_2$ using 2's complement. 1100

- (b) Perform $(715)_8 - (237)_8$ 456

- B. Convert $(1427)_8$ into hexa-decimal number. $(001100010111) = 317$ 2

- C. (i) Convert $(1110011011)_2$ number into octal number. 1633 3

- (ii) Convert $(ADE7)_{16}$ number into decimal number. 44519



$$\alpha = \frac{B}{1+B}$$

Gain is less affected by noise
Small size
more versatile
true in memo

[P.T.O]

16
11
2

Q-6 A. Derive the relationship between α and β .

B. Name the three possible transistor configurations. Also show that α is always less than unity.

C. Describe the transistor action in detail.

OR

Q-6 A. (i) Give in brief about operating point.

(ii) calculate emitter current in a transistor for which $\beta = 40$ and base current $= 15\mu A$
(iii) find the α if (a) $\beta=90$; (b) $\beta=40$

(iii) find the α if (a) $\beta=90^\circ$; (b) $\beta=40^\circ$
Explain.

B. Explain common emitter configuration in detail and also derive the useful equations.

C. For the circuit shown in figure, draw the DC load line

